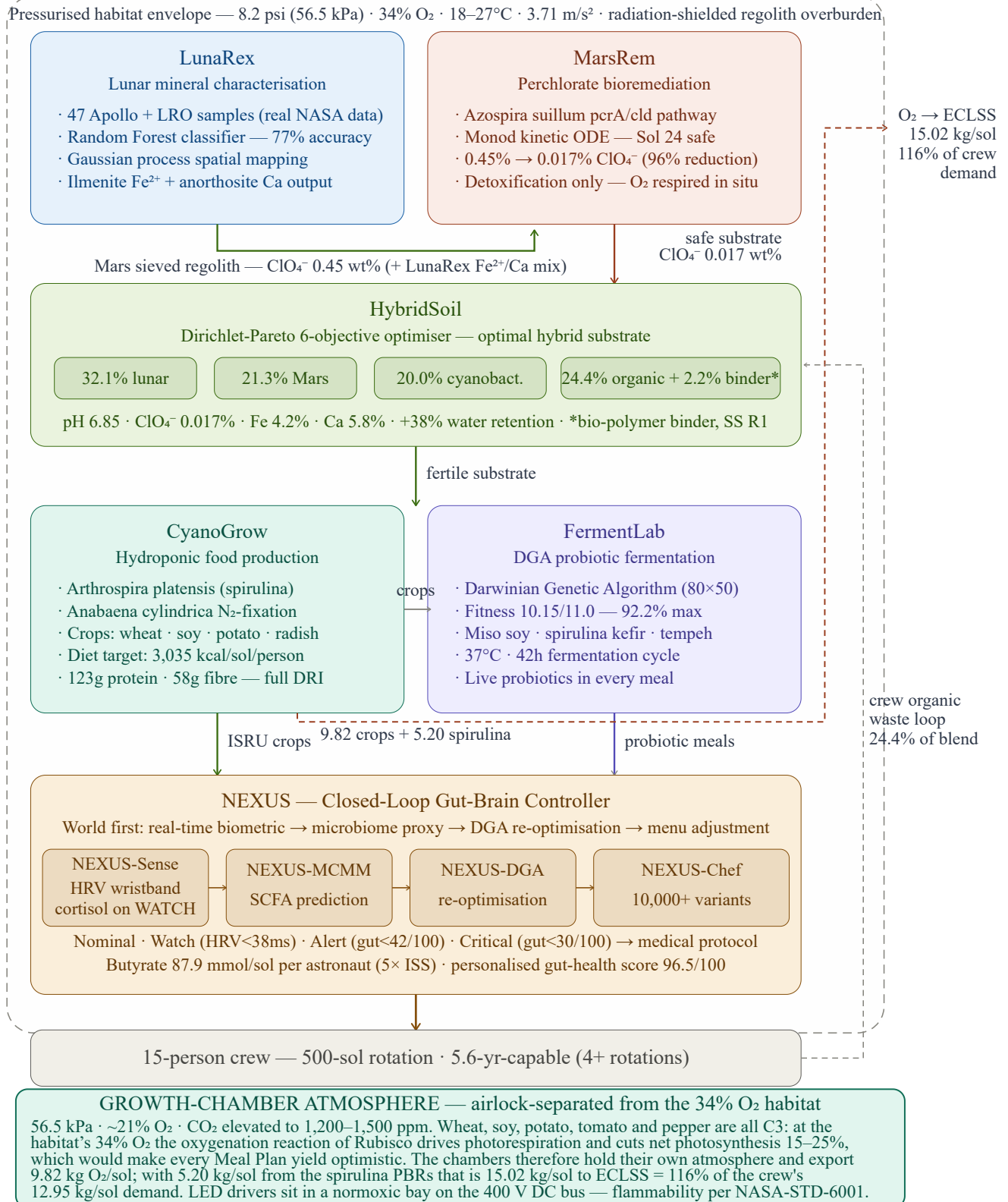


ESPERANZA II — Habitat Design Layout

NASA Mars to Table Challenge 2026 · Team ESPERANZA Research · María Jesús Puerta Angulo



Resource flows: —→ substrate / nutrients —→ O₂ / ECLSS —→ biometric loop - - - -> closed-loop return

Modules: LunaRex MarsRem HybridSoil CyanoGrow FermentLab NEXUS

511 m² growing area = 373 LED canopy + 40 spirulina + 48 insect + 50 mushroom · LED sole-source, 3.0 μmol/J
Power 58.1 kW mean · 64.0 kW peak · 77 kW installed (20% margin) = 10.7 process + 30.4 LED + 5.0 canopy HVAC + 2.2 dehumid
+ 2.5 thermal loop + 2.5 nutrient pumps + 1.2 pasteurisation + 0.8 cold room + 0.8 task lighting/crop automation + 0.7 galley
+ 0.6 mushroom + 0.4 water/UV + 0.3 CO₂ · LED 30.4 kW mean / 33.3 kW peak (+9%): wheat USU-Apogee 24/0 baseload 13.52 kW,
soy 13.22 + potato 12.02 kW on opposite 12.33 h windows · ~300 m² radiator: N–S vertical panels, 40 °C loop, low-α face
Energy storage 100 kWh = 25 kW × 4 h · 10 m², 25 m³ and ~500 kg are habitat-power scope, outside the 1,058 m³ envelope
Automation: rail gantry + manipulator arm + mobile trays — crop production, harvest and replanting 22.0 h per 5-sol cycle
644 L/sol transpiration · 9.7 L/sol make-up · 47.0% Earth-provisioned · 1,058 m³ pressurised — schedule on Sheet 2
47.0% Earth-provisioned · 45.4% with the in-situ EVA supplement (sensitivity: +25 m² canopy, +2.0 kW — App. §9)
Crew time: specialists 81.2 of 90 h per 5-sol (90%) · worst specialist 40.6 of 45 h · 0 of 10 windows exceed the limit

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Internal engineering model; parameters in ConOps Appendix A8 · NASA Mars to Table Challenge 2026

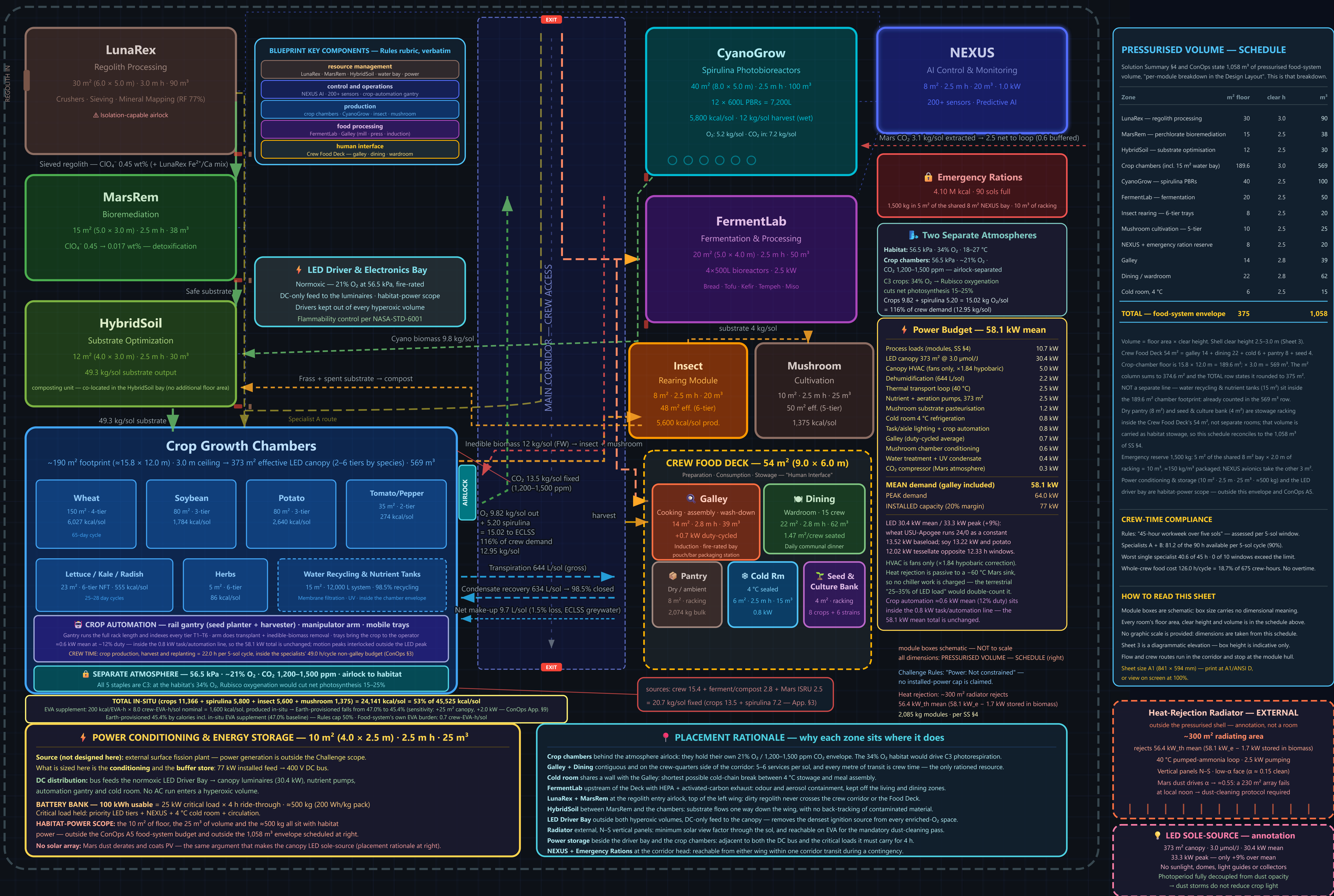
ESPERANZA II — Habitat Food System Blueprint

NASA Mars to Table Challenge 2026

15 Crew · 500 Sols · 511 m² Growing Area (373 m² LED canopy + 98 m² heterotrophic + 40 m² spirulina) · 1,058 m³ Total Volume · LED sole-source · 58.1 kW mean / 64.0 kW peak / 77 kW installed · 100 kWh battery bank (4 h critical ride-through) · rail-gantry crop automation · crop chambers on a separate 21% O₂ atmosphere

TOP-DOWN FLOOR PLAN

HABITAT OUTER SHELL — PRESSURIZED ENVELOPE



Mars Atmosphere — 95% CO₂ · 6.1 mbar

REGOLITH OVERBURDEN — RADIATION SHIELDING

Continuous over the pressurised shell — the envelope declared on Sheet 1
Overburden depth is habitat-structures scope and is not sized in this submission

EXTERNAL RADIATOR — ~300 m² · N-S vertical panels · 40 °C loop
rejects 56.4 kW_{th} mean (58.1 kW_e – 1.7 kW stored in biomass) · low-α face + dust-cleaning

PRESSURIZED HABITAT SHELL — 2.5–3.0 m ceiling height

40 °C pumped-ammonia feedthrough
waste heat ↑ 56.4 kW_{th}

CROP GROWTH CHAMBERS

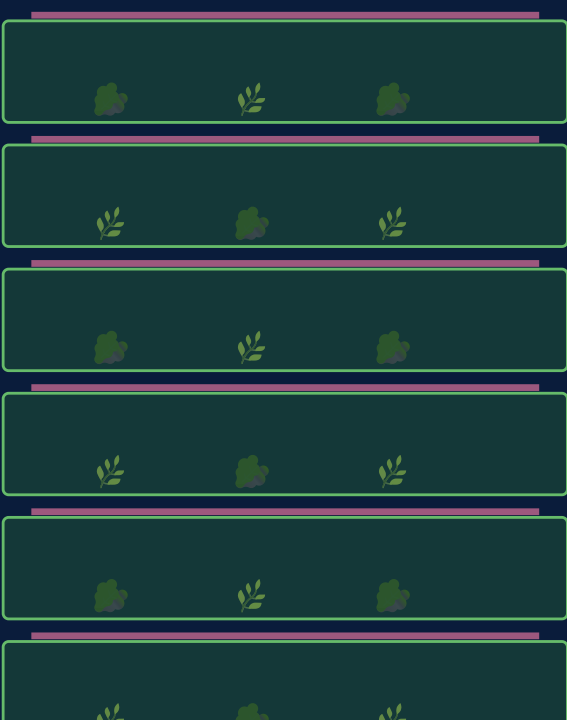
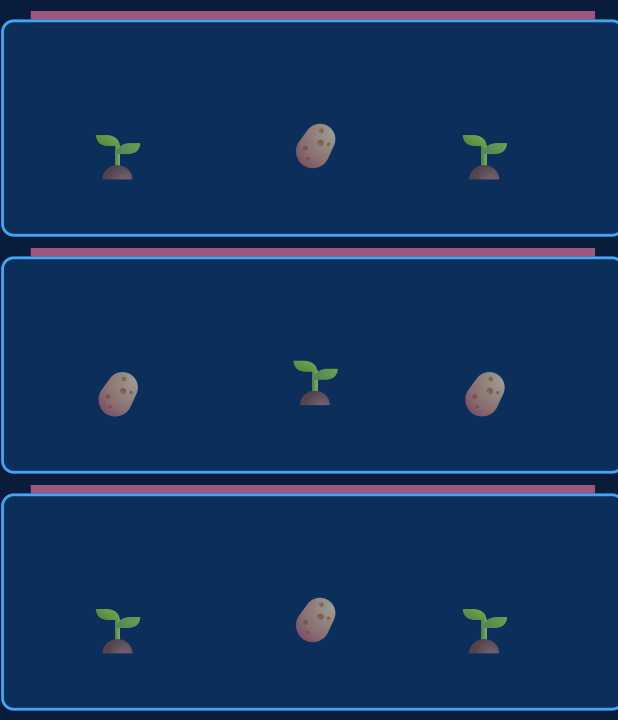
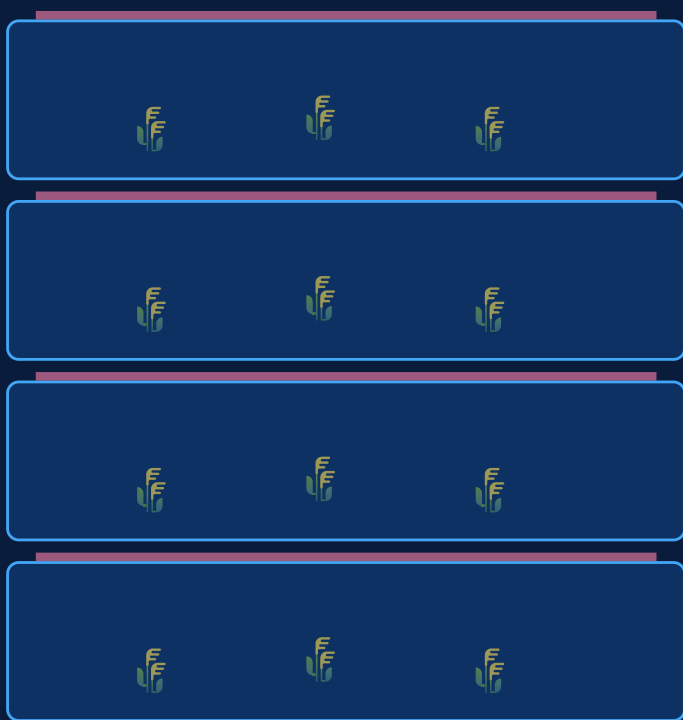
190 m² footprint (113 m² racks + 62 m² aisles + 15 m² water bay) → 373 m² LED canopy | 569 m³ | LED sole-source

🔒 SEPARATE ATMOSPHERE — 56.5 kPa · ~21% O₂ · CO₂ 1,200–1,500 ppm · airlock to the 34% O₂ habitat

Wheat (4-tier)

Soy + Potato (3-tier)

Leafy/Herbs (6-tier NFT)



🤖 SOWING / HARVEST GANTRY ON RAILS — traverses the full rack length, indexes every tier

Manipulator arm: transplant + removal of inedible biomass · mobile trays deliver the crop to the operator instead of the operator to the crop
≈0.6 kW mean (12% duty), inside the 0.8 kW task/automation line — 58.1 kW unchanged · crop tending and harvest 22.0 h per 5-sol cycle

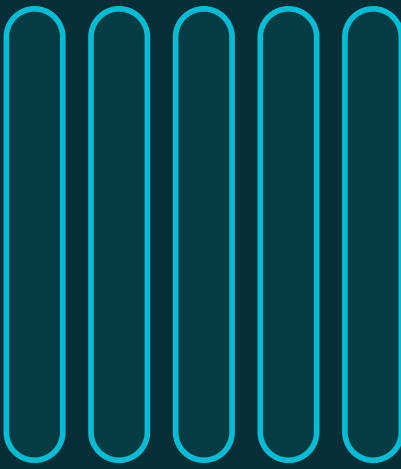
MAINTENANCE CORRIDOR — 1.8 m clearance



AIRLOCK

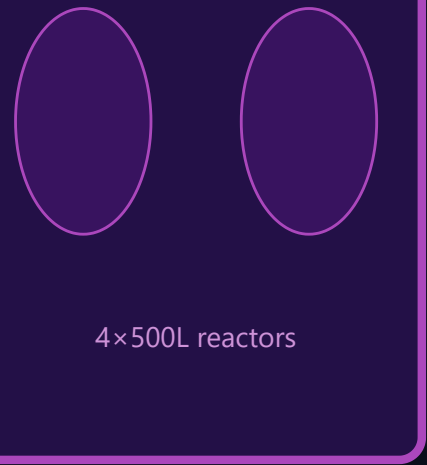
CyanoGrow

PBR Array · 2.5 m tall



FermentLab

2.5 m



Insect Module

6-tier stacked trays · 2.5 m

🪱 Mealworms (*T. molitor*)

NEXUS

AI Control · 2.5 m

200+ sensors

Mushroom

5-tier shelving · 2.5 m

🍄 Oyster + Shiitake

Galley

14 m² = 4.0 × 3.5 m · 2.8 m h

Induction · mill · wash-down
fire-rated bay

Dining

22 m² = 5.5 × 4.0 m · 15 crew

Wardroom table · 2.8 m

Pantry 8 · Cold 6 m²

in-deck stowage racking

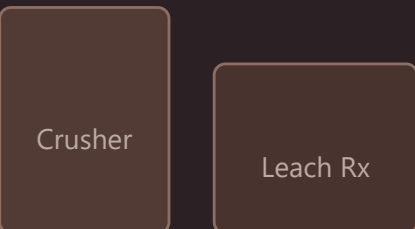
— = LED grow lights, 3.0 μmol/J
SOLE-SOURCE — no natural light, no
daylight path, no light collectors
373 m² canopy · 30.4 kW mean / 33.3 kW peak

LED Driver & Electronics Bay

Normoxic 21% O₂ · fire-rated
DC distribution only to luminaires
NASA-STD-6001 flammability

LunaRex

3.0 m height



MarsRem

2.5 m



HybridSoil

12 m² · 2.5 m

Mixing + Dosing

NASA-STD-3001 Vol.2 §7 · HUMAN FACTORS & REDUNDANCY & HACCP & SCALABILITY

Human factors: wardroom seats all 15 crew at one sitting, 1.47 m²/crew, daily communal dinner (Sheet 2) · 1.8 m maintenance clearance corridor (above)
Redundancy: seed & culture bank, 8 crops + 6 strains (Sheet 2) · 100 kWh battery = 4 h critical-load ride-through · emergency rations 4.10 M kcal = 90 sols full
Isolation: crop atmosphere isolable behind its airlock (Sheets 2–3) · isolation doors on each wing bay · LunaRex isolation-capable airlock (Sheet 2)
HACCP: crop-tissue perchlorate is a declared CCP · 10-CCP re-validation at handover + per-cycle HACCP audit (Solution Summary) · HACCP + water monitoring 9.0 h/cycle (ConOps App. §5)
Food safety: fermentation/crop bays in fire-rated enclosures with O₂ trim per NASA-STD-3001 Vol.2 §6.4 (SS §4) · galley fire-rated induction bay ·
FermentLab HEPA + activated-carbon exhaust · 4 °C cold room wall-shared with the galley (Sheet 2)
Scalability: 4+ consecutive 500-sol crew rotations (≈5.6 Earth years) with 50-sol hand-over (SS §5b) · crew time 81.2 of the 90 h per 5-sol cycle — margin retained (Sheet 2)

annotation panel — not a sub-floor volume

⚡ POWER CONDITIONING & ENERGY STORAGE — 10 m² (4.0 × 2.5 m) · 2.5 m h · 25 m³

Source: external surface fission plant — generation is out of Challenge scope and is not designed here.
Conditioned here: 77 kW installed → 400 V DC bus → LED driver bay, pumps, automation, cold room.

Battery bank 100 kWh = 25 kW critical × 4 h ride-through · ≈500 kg · 200 Wh/kg pack level
Holds priority LED tiers + NEXUS + 4 °C cold room + circulation. No solar array: dust derates PV.
Habitat-power scope: the 10 m², 25 m³ and ≈500 kg sit outside ConOps A5 and the 1,058 m³ envelope.

annotation panel — not a sub-floor volume

Regolith in

Sheet size A1 (841 × 594 mm) — print at A1/ANSI D, or view on screen at 100%

MARS SURFACE — REGOLITH

DIAGRAMMATIC SECTION — NOT TO SCALE · box height is indicative only: modules that lie behind one another in plan are drawn stacked for legibility · every clear height is called out here and scheduled on Sheet 2

LEGEND

■ LunaRex (Regolith) ■ MarsRem (Bioremediation) ■ HybridSoil (Substrate) ■ CyanoGrow (Spirulina) ■ FermentLab (Fermentation) ■ NEXUS (AI Control) ■ Crop Growth Chambers ■ Insect Rearing ■ Mushroom Cultivation ■ Emergency Rations ■ Galley (cooking / meal assembly) ■ Dining / Wardroom (15 crew) ■ Dry Pantry ■ Cold Room 4 °C ■ Seed & Culture Bank ■ Heat-Rejection Radiator (~300 m³)
■ Crop-chamber atmosphere (21% O₂, CO₂ 1,200–1,500 ppm) ■ Atmosphere airlock / gas interface ■ LED Driver & Electronics Bay (normoxic) ■ Power Conditioning & Energy Storage (100 kWh, 4 h) ■ Crop Automation (gantry · arm · mobile trays)

Legend applies to Sheets 2–3

FLOW-LINE KEY — Sheet 2 routes: - - - - Water — transpiration 644 L/sol · condensate 634 L/sol · make-up 9.7 L/sol — Substrate / nutrient — clean → safe → optimised substrate 49.3 kg/sol · cyano biomass 9.8 kg/sol (dashed) - - - - O₂ export — crops 9.82 + spirulina 5.20 = 15.02 kg/sol to ECLSS (116% of crew demand) - - - - Waste / biomass — inedible 12 kg/sol (FW) → insect + mushroom · substrate 4 kg/sol · frass + spent → compost
- - - - CO₂ / air — 13.5 kg/sol fixed by crops (1,200–1,500 ppm) · Mars CO₂ make-up 3.1 kg/sol extracted → 2.5 net to loop (0.6 buffered) — Harvest — crop chambers → Galley - - - - Specialist A route — corridor → left-wing modules → crop chambers - - - - Specialist B route → Galley → Dining ····· NEXUS sensor network — telemetry trunk to all modules - - - - Main corridor — crew access spine ■ Emergency Exit / Isolation Door
— LED Grow Lights (sole-source, 3.0 μmol/J)